Temporal Frame Ordering

Transformer Ranking

MLWARE-26 Sherlock Files - Decoding Motion Causality

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The Problem: Sherlock Files

Reconstructing precise chronological timelines from intentionally scrambled video fragments, relying entirely on physical cues and visual causality.

The Big Idea: Ranking, Not Classification

- **Not a binary problem:** We are not labeling isolated frames as simply "early" or "late."
- **Inherent Relational Logic:** To rank elements, you must compare them. You can only deduce "Frame A precedes Frame B" by evaluating them simultaneously.
- **MLP Limitations:** Standard Multi-Layer Perceptrons score frames independently, failing to capture vital relative temporal differences.
- **The Goal:** Given N shuffled frames, output a permutation that definitively orders them according to the unbroken laws of physics.

Key Competition Metrics

0.274

BEST VAL KENDALL'S TAU

211

TRAINING EPOCHS

Understanding Kendall's Tau (τ)

This metric measures how accurately our predicted sequence aligns with the ground truth by counting concordant and discordant frame pairs.

A Score of +1: Indicates a perfectly reconstructed chronological timeline.

A Score near 0: Suggests random guessing. Our model's score of 0.2741 represents significant, verifiable learning over baseline.

The model initiated early stopping at epoch 211 after a 60-epoch patience window, indicating maximum extraction of signal from the provided features.

The 5-Stage Pipeline

1. Extract & Feature

Sample up to 64 evenly-spaced frames per video. Pass each through a frozen CLIP ViT-B/16 model to generate 512-dimensional visual embeddings.

2. Physics & Model

Compute 8 hand-crafted physics signals per frame. Feed all features into a 4-layer Transformer Encoder for powerful cross-frame self-attention.

3. Inference & TTA

Score each frame and execute 3 Test-Time Augmentation runs (normal, reversed, noisy). Average scores and argsort to yield the final prediction.

Feature Extraction Strategy

Frozen CLIP ViT-B/16

We leverage OpenAI's Contrastive Language-Image Pre-training model. By caching these 512-dimensional embeddings, we save vast computational resources.

Why Frozen? Fine-tuning updates memory weights without updating the cached disk features, creating a lethal mismatch. Freezing ensures consistency.

Why B/16? It processes 196 spatial patches per image (vs 49 in B/32), capturing the fine spatial detail crucial for analyzing subtle motion.

Hand-Crafted Physics Signals

Neural networks require massive datasets to intuit basic physics. We bypass this by feeding explicit, low-level physical observations alongside CLIP features.

Centre-of-mass (X,Y): Tracks objects falling or traversing the frame.

Sharpness: Detects motion blur at points of impact.

Entropy & Edges: Measures structural complexity and debris scatter.

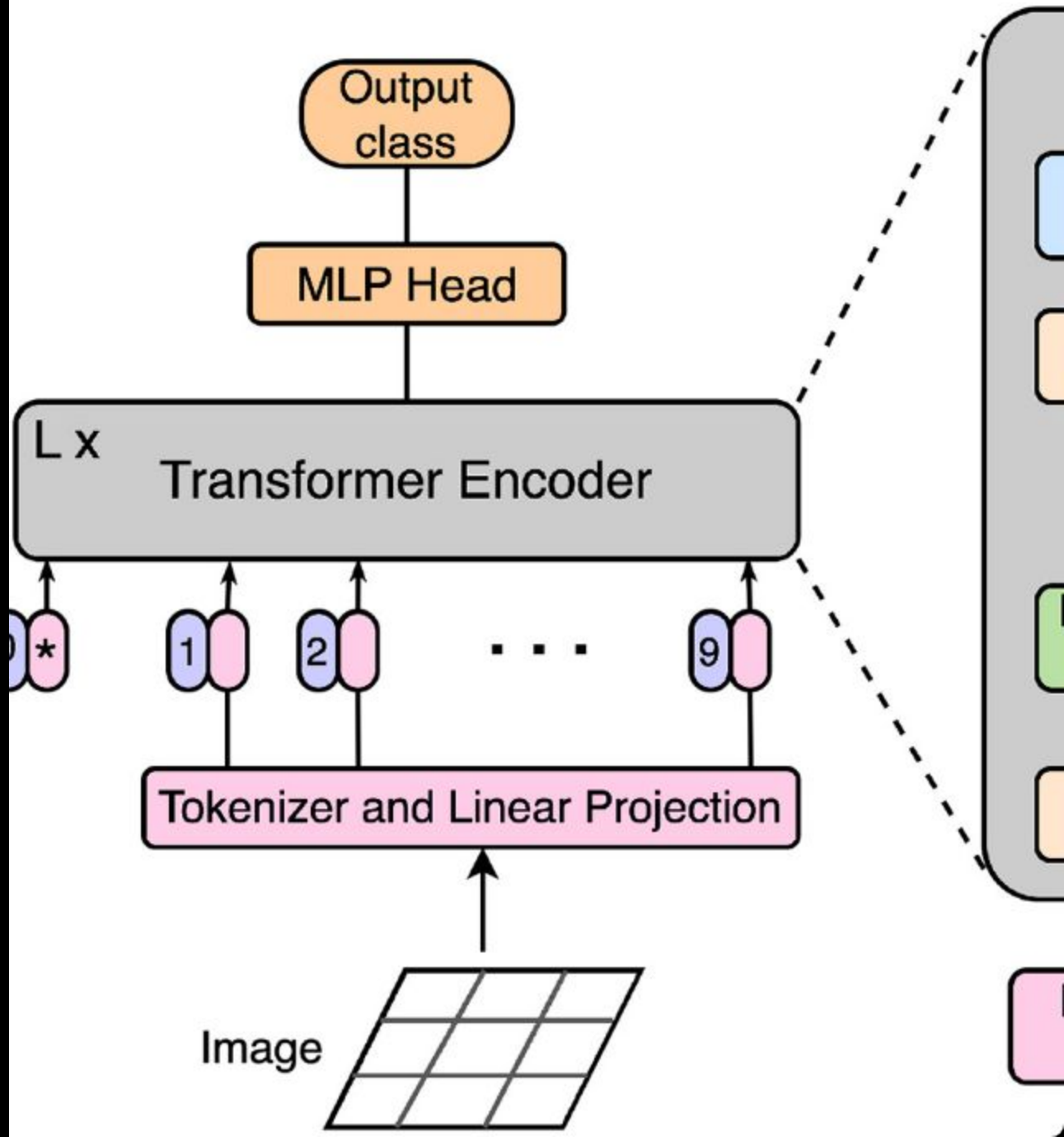
Pixel-level Motion: Quantifies absolute inter-frame changes.

Why a Transformer?

Earlier iterations relied on Multi-Layer Perceptrons (MLPs), which scored frames in isolation against a single global average. This failed to capture dynamic causality.

The Solution: A 4-layer Transformer Encoder.

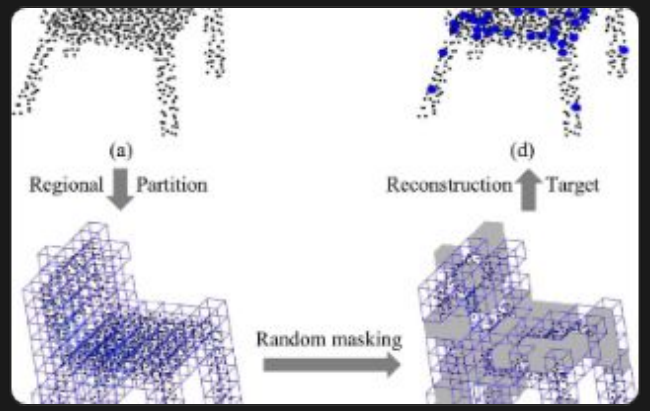
Self-attention computes a weighted interaction matrix across all frames. Every single frame "asks a question" (Query) and evaluates all other frames (Keys/Values). This allows the model to deduce logic such as: "Frame A shows the object at peak height, thus it strictly precedes Frame B where it descends."



Core Innovations: Bridging Train/Test Gaps

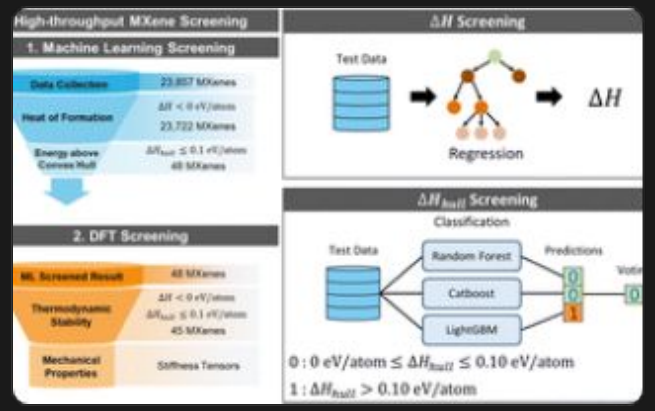
No Positional Encoding

Training data is cached sequentially. Standard positional encodings allowed the model to cheat. Removing it forces pure content-based ordering.



Sequence Shuffle

Applied to 100% of training steps. By scrambling sequences before feeding them in, the model can no longer rely on sequence placement to predict time.



Reversal & Noise

Temporal reversal (30%) teaches symmetry. CLIP feature noise (50%) prevents overfitting to exact numerical embeddings, ensuring robust inference.

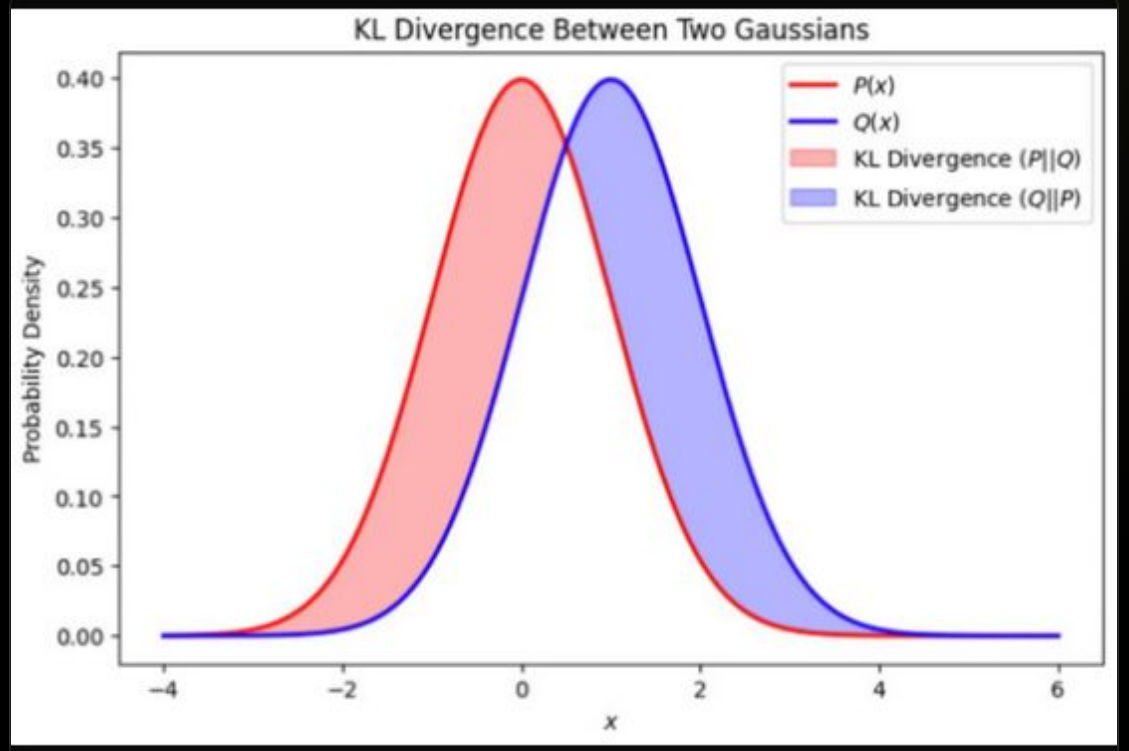
Distance-Weighted Pairwise BCE

We approach ordering by asking: "Does frame *j* occur after frame *i*?" using Pairwise Binary Cross-Entropy.

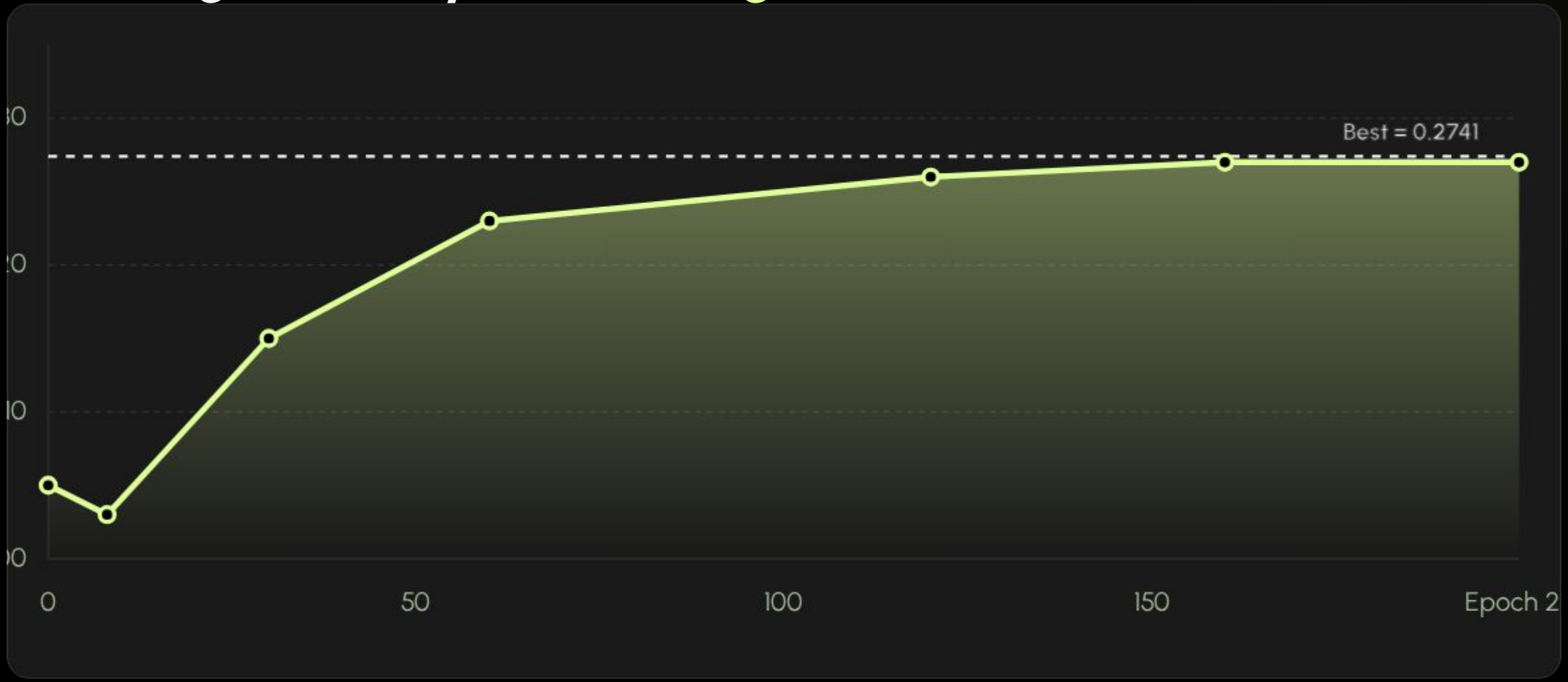
The breakthrough innovation lies in **Distance Weighting**.

Frames separated by wide temporal gaps are weighted heavily. The model prioritizes obvious global ordering (start vs. end) over agonizing over nearly identical adjacent frames.

$$\text{Loss} = \frac{\sum (\text{BCE} \times w)}{\sum w}, \quad w = |r_j - r_i|$$



Training Stability & Convergence



Validation Kendall's Tau: Rapid learning observed through Epoch 60, followed by careful refinement and an eventual plateau near 0.274. No signs of overfitting (falling validation) were present, validating our rigorous data augmentation.

Robust Inference with Test-Time Augmentation

Run 1: Normal Forward Pass. Standard scoring executed on the original, unmodified frame sequence presented during testing.

Run 2: Reversed Sequence. The sequence is flipped, scored, and inverted. Because the model was trained on reversal augmentation, it understands symmetry. Averaging this run cancels out unintended asymmetric model biases.

Run 3: Jittered CLIP Noise. Applying slight Gaussian noise (std=0.01) to features smooths the uncertainty around borderline decision boundaries where temporal adjacency is visually ambiguous.

Final Synthesis: The scores from all three runs are averaged. We execute a final argsort to transform these scores into the strict chronological ranking required for submission.

Case Solved.

The true timeline of events has been reconstructed.

MLWARE-26 · Sherlock Files Technical Report
Temporal Frame Ordering via Relational Transformers

Image Sources

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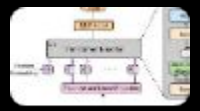

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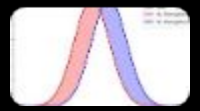

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